

# RVCaIPy radial-velocity benchmark report

Broadening function: LL Aqr and AK For · Cross-correlation: 16 UVES high-velocity stars  
RVCaIPy, <https://github.com/atemelci/RVCaIPy> · 2026 September 22

RVCaIPy velocities were compared with published measurements of the same spectra or the same nights in three independent tests. With the **broadening function**, the HARPS velocities of LL Aqr are reproduced to 0.092 and 0.142 km s<sup>-1</sup> (primary, secondary) with reduced  $\chi^2$  of 1.03 and 0.81; for AK For the primary behaves the same way (0.150 km s<sup>-1</sup>), while the secondary shows a velocity-amplitude deficit of 0.8% traced to template mismatch. With **cross-correlation**, the 16 UVES velocities of Alazzawi et al. (2026) are reproduced to 0.49 km s<sup>-1</sup> in the median, with a velocity-scale error of  $+0.017 \pm 0.048\%$  over a 919 km s<sup>-1</sup> baseline.

## 1. What was tested

|                     | LL Aqr                                        | AK For                                        | UVES sample                                               |
|---------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| Method              | Broadening function                           | Broadening function                           | Cross-correlation                                         |
| Reference           | Graczyk et al. (2016), Table 1                | Helminiak et al. (2014), Table 2              | Alazzawi et al. (2026), Table 3, $V_r^{\text{meas}}$      |
| Instrument          | HARPS                                         | FEROS (14) + HARPS (2)                        | UVES, DIC1 390+580, red arm                               |
| Measured / compared | 17 / 16 nights                                | 16 / 14 nights                                | 16 / 16 stars, one epoch each                             |
| Template            | HIP 57757 (6093 K)                            | HIP 108870 (4676 K)                           | synthetic, 5215 K, log g 3.55                             |
| Window / step       | 500–550 nm, $\Delta v = 1$ km s <sup>-1</sup> | 500–550 nm, $\Delta v = 2$ km s <sup>-1</sup> | 500–550 nm                                                |
| What it tests       | SB2 amplitudes, 20.18 d orbit                 | SB2 amplitudes, 3.98 d orbit, cool secondary  | zero point and velocity scale over 919 km s <sup>-1</sup> |

**Table 1.** Material and settings. All spectra are public ESO archive phase-3 products; only epochs present in both data sets are compared.

## 2. How the numbers are defined

For each matched epoch the difference between our velocity and the published one is formed, with the two uncertainties added in quadrature.  $\Delta$  is the weighted mean offset (a zero point, absorbed by the systemic velocity in any orbit fit),  $s$  the scatter,  $\chi_r^2$  the test of whether that scatter is explained by the quoted errors, and the slope  $a$  of the straight-line fit measures an amplitude or velocity-scale error ( $a \neq 1$ ), as distinct from a zero point ( $b \neq 0$ ):

$$d_i = v_i^{\text{ours}} - v_i^{\text{pub}}, \quad \sigma_i = \sqrt{\varepsilon_i^{2, \text{ours}} + \varepsilon_i^{2, \text{pub}}}, \quad w_i = \sigma_i^{-2}$$

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}, \quad s = \sqrt{\frac{\sum (d_i - \Delta)^2}{N-1}}, \quad \chi_r^2 = \frac{1}{N-1} \sum \left( \frac{d_i - \Delta}{\sigma_i} \right)^2, \quad v^{\text{ours}} = a v^{\text{pub}} + b$$

**No  $\chi^2$  for the UVES test.** There our velocities and the published ones come from the *same exposures*, so the  $\sim 0.5$  km s<sup>-1</sup> slit-centring error that dominates the published budget is common to both and cancels in the difference, and no per-star error is published. The scatter is then reported as what it is — the method-to-method disagreement — without a  $\chi_r^2$ .

## 3. Broadening function: LL Aqr and AK For

|                                       | LL Aqr             |                    | AK For             |                    |
|---------------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                       | Primary            | Secondary          | Primary            | Secondary          |
| Nights compared                       | 16                 | 16                 | 14                 | 14                 |
| Offset $\Delta$ (km s <sup>-1</sup> ) | $+0.251 \pm 0.023$ | $+0.294 \pm 0.040$ | $+0.208 \pm 0.059$ | $+0.318 \pm 0.130$ |
| Scatter $s$ (km s <sup>-1</sup> )     | 0.092              | 0.142              | 0.150              | 0.864              |
| Reduced $\chi^2$                      | 1.03               | 0.81               | 0.45               | 2.54               |
| Slope $a$                             | 0.9997             | 0.9989             | 0.9986             | 0.9923             |
| Amplitude error                       | -0.03%             | -0.11%             | -0.14%             | -0.77%             |
| Authors' own fit rms                  | 0.053              | 0.049              | 0.082              | 0.096              |

**Table 2.** Broadening-function velocities against the published ones. The last row is the residual rms of the authors' own orbital fits, for scale.

**LL Aqr — passes.** After offsets of  $+0.251$  and  $+0.294$  km s<sup>-1</sup> the residuals scatter by 0.092 and 0.142 km s<sup>-1</sup> with  $\chi_r^2$  near unity: the differences are fully explained by the quoted errors. Amplitudes agree to better than 0.15%, so an orbit fitted to our velocities returns the same masses.

**AK For — primary passes, secondary biased.** The primary behaves like LL Aqr ( $0.150 \text{ km s}^{-1}$ ,  $\chi^2_r = 0.45$ ). The secondary gives  $\chi^2_r = 2.54$ , its residuals running from  $-1.5 \text{ km s}^{-1}$  near phase 0.1 to  $+1.3 \text{ km s}^{-1}$  beyond 0.6 — a 0.8% amplitude deficit, not noise. The single 4676 K template is  $\sim 300 \text{ K}$  hotter than the 4390 K secondary; since  $K_2$  sets  $M_1$ , an orbit from our velocities alone would underestimate  $M_1$  by about one per cent. A  $\sim 4400 \text{ K}$  template for the secondary is the obvious retest.

#### 4. Cross-correlation: 16 UVES high-velocity stars

| Star             | RV, this work<br>[ $\text{km s}^{-1}$ ] | V_r meas<br>[ $\text{km s}^{-1}$ ] | V_r Gaia DR3<br>[ $\text{km s}^{-1}$ ] | $\Delta$<br>[ $\text{km s}^{-1}$ ] |
|------------------|-----------------------------------------|------------------------------------|----------------------------------------|------------------------------------|
| TYC 7524-00065-1 | $+351.498 \pm 0.027$                    | $+351.920$                         | $+350.953 \pm 0.217$                   | $-0.422$                           |
| TYC 7535-00160-1 | $+355.074 \pm 0.037$                    | $+353.523$                         | $+355.069 \pm 0.253$                   | $+1.551$                           |
| C0213360-505024  | $+387.999 \pm 0.027$                    | $+388.600$                         | $+388.222 \pm 0.611$                   | $-0.601$                           |
| C1012254-203007  | $+508.139 \pm 0.035$                    | $+506.755$                         | $+508.138 \pm 1.068$                   | $+1.384$                           |
| C1032508-244851  | $+509.755 \pm 0.029$                    | $+509.490$                         | $+515.354 \pm 2.470$                   | $+0.265$                           |
| C1126364-293415  | $+411.154 \pm 0.019$                    | $+409.870$                         | $+409.674 \pm 0.926$                   | $+1.284$                           |
| C1250398-030748  | $+398.245 \pm 0.031$                    | $+403.020$                         | $+396.221 \pm 0.979$                   | $-4.775$                           |
| C1302091-323721  | $+410.550 \pm 0.021$                    | $+410.000$                         | $+409.817 \pm 3.027$                   | $+0.550$                           |
| C1304064-331913  | $+360.461 \pm 0.019$                    | $+360.720$                         | $+357.825 \pm 0.994$                   | $-0.259$                           |
| C1330590-230955  | $+335.542 \pm 0.027$                    | $+335.350$                         | $+334.919 \pm 0.526$                   | $+0.192$                           |
| TYC 7274-00734-1 | $+408.425 \pm 0.020$                    | $+408.317$                         | $+407.955 \pm 0.208$                   | $+0.108$                           |
| C1519196-191359  | $-409.410 \pm 0.026$                    | $-409.610$                         | $-409.694 \pm 6.570$                   | $+0.200$                           |
| C1536201-144228  | $-329.600 \pm 0.023$                    | $-329.600$                         | $-328.647 \pm 2.112$                   | $-0.000$                           |
| C1919566-632839  | $+330.810 \pm 0.032$                    | $+328.680$                         | $+330.399 \pm 0.530$                   | $+2.130$                           |
| C2136142-694908  | $+383.373 \pm 0.017$                    | $+381.640$                         | $+382.882 \pm 0.330$                   | $+1.733$                           |
| TYC 8019-00159-1 | $+99.097 \pm 0.035$                     | $+99.117$                          | $+98.450 \pm 0.171$                    | $-0.020$                           |

**Table 3.** Row for row in the order of Table 3 of Alazzawi et al. (2026). Our velocities are barycentric;  $\pm$  is the formal error of the Gaussian fit to the CCF peak.  $V_r^{\text{meas}}$  is their template-matching velocity from the same spectrum (no per-star error published), and  $\Delta = \text{ours} - V_r^{\text{meas}}$ .  $V_r^{\text{Gaia DR3}}$  is listed for reference only and enters no statistic.

|                                                 | All 16 stars            | Excluding C1250398-030748 |
|-------------------------------------------------|-------------------------|---------------------------|
| Weighted offset $\Delta$ ( $\text{km s}^{-1}$ ) | $+0.207 \pm 0.125$      | $+0.539 \pm 0.129$        |
| Median $ \Delta $ ( $\text{km s}^{-1}$ )        | 0.486                   | 0.422                     |
| Scatter $s$ ( $\text{km s}^{-1}$ )              | 1.564                   | 0.854                     |
| Within $1 \text{ km s}^{-1}$                    | 10 of 16                | 10 of 15                  |
| Slope $a$                                       | $1.000171 \pm 0.000479$ | $1.000771 \pm 0.000482$   |
| Velocity-scale error $a - 1$                    | $+0.017 \pm 0.048\%$    | $+0.077 \pm 0.048\%$      |

**Table 4.** Summary of Table 3, with and without the single outlier so its effect on each statistic is visible.

**Velocity scale and zero point — pass.** Over  $919 \text{ km s}^{-1}$  the slope is consistent with unity to 0.05%, so nothing in the pipeline (wavelength solution,  $\log\lambda$  resampling, barycentric correction) stretches the velocity axis. The offset of  $+0.207 \text{ km s}^{-1}$  is inside the  $\sim 0.5 \text{ km s}^{-1}$  the paper attributes to slit centring.

**Star-to-star agreement at the template-mismatch level.** 10 of 16 stars agree within  $1 \text{ km s}^{-1}$ , and the scatter is  $0.85 \text{ km s}^{-1}$  without the outlier — the price of one solar-scaled 5215 K template over 500–550 nm against per-star matched templates over the full red arm, for stars of 4000–5486 K and  $[\text{Fe}/\text{H}] -1.2$  to  $-2.9$ .

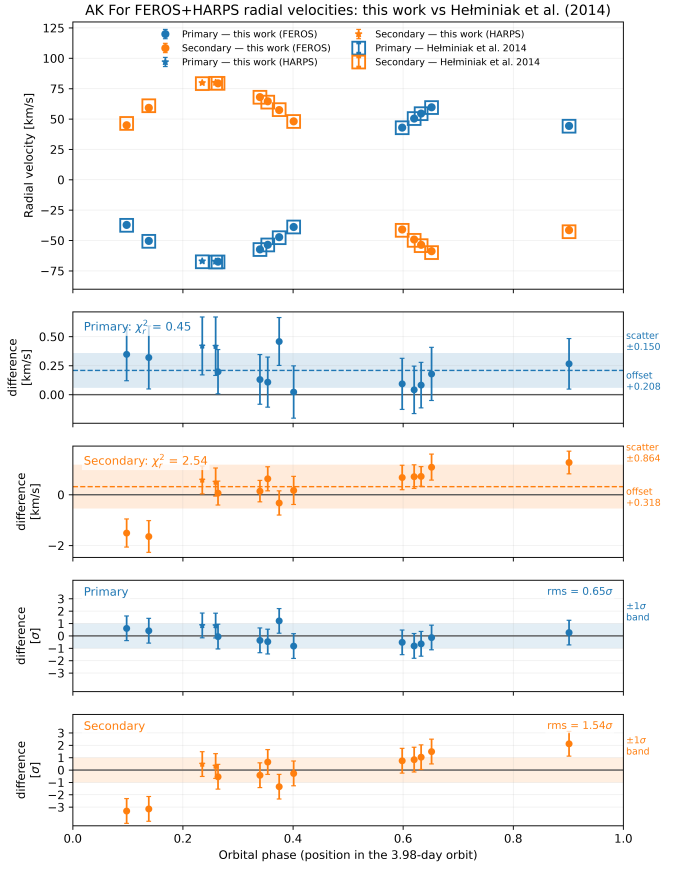
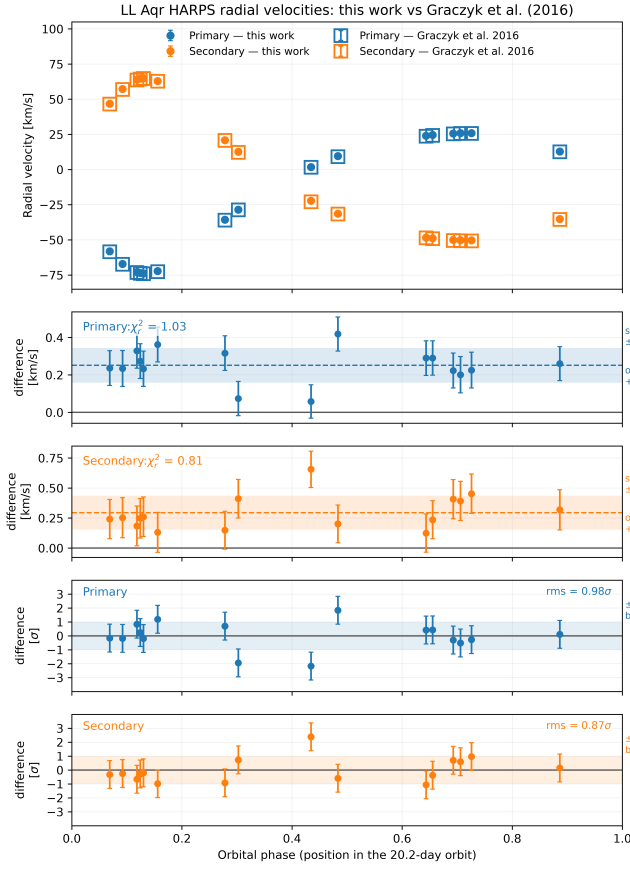
**One unexplained outlier.** C1250398-030748 differs by  $-4.775 \text{ km s}^{-1}$ . Nothing in our run marks it out (S/N 161, single symmetric CCF peak of significance 31.8, and at 5443 K one of the best template matches); it is kept, and a matched-template retest is pending.

#### 5. Limits

The binaries test amplitudes but not the absolute zero point; the UVES sample tests the zero point and scale but, with one epoch per star and single CCF peaks, not amplitudes or blended profiles. Both use a single template per data set, which is the pessimistic case — the AK For secondary bias and the UVES scatter both trace back to it. The UVES run was configured with  $R = 80000$  against the delivered  $R \approx 40\,000$ , which widens nothing but the template broadening kernel. Finally, the UVES comparison shares the archive wavelength calibration with the reference, so it shows that two codes agree on the same data, not that either is right in an absolute sense.

#### References

- Alazzawi, J. M. K., Caffau, E., Bonifacio, P., et al. 2026, A&A, DOI 10.1051/0004-6361/202659789  
Graczyk, D., Smolec, R., Gazeas, K., et al. 2016, A&A, 594, A92  
Hełminiak, K. G., Graczyk, D., Konacki, M., et al. 2014, A&A, 567, A64  
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**Figure 1.** Broadening-function velocities against the published ones: LL Aqr versus Graczyk et al. 2016 (left, 16 nights) and AK For versus Helminiak et al. 2014 (right, 14 nights; circles FEROS, stars HARPS). Top: phased velocity curves (filled: this work, open: published). Middle: differences with the weighted offset (dashed) and scatter (band). Bottom: residuals in units of their own  $1\sigma$  after removing the offset. The AK For secondary residuals slope with phase — the signature of a too-small amplitude.